Diagnosing Direct-Substitution and One-Sided-Sign Errors Answer Key

AP Calculus AB / BC — Limits and Continuity

Quick Answers

1. 4	4. $-\infty$, ∞	7. 5, 14
2. 4, 4	5. -1, 1	8. —
3. $\frac{1}{4}$, 0.25	6. ∞, $-\infty$	

Curriculum

Level: AP Calculus AB / BC — Limits and Continuity
LIM-1, LIM-2 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 14 tagged checks.

Common wrong answers

2. *If they got 2: cancelled the common factor to x - 1 instead of x + 1*
3. *If they got 0.5: kept only the square root in the rationalized denominator instead of the whole conjugate sum*
7. *If they got 23: read k straight off the right-hand branch value instead of solving 9 + k = 14, which leaves the left-hand limit at 23*

1. $\lim_{x\to 3}(x^2 - 4x + 7)$.

Solution: A polynomial is continuous everywhere, so the limit equals the value: $3^2 - 4(3) + 7 = 9 - 12 + 7$. $\boxed{4}$

That is the whole content of "direct substitution works here": continuity at the point is the licence, and it is worth naming rather than assuming.

2. $\lim_{x\to 3}\dfrac{x^2 - 2x - 3}{x - 3}$.

Solution: Substitution first, so the diagnosis is on the page: $\dfrac{9-6-3}{3-3} = \dfrac{0}{0}$ — indeterminate, so the substitution told us only that $x - 3$ divides both parts. Factor and cancel:

$$\frac{x^2 - 2x - 3}{x - 3} = \frac{(x-3)(x+1)}{x-3} = x + 1 \qquad (x \neq 3)$$

Now substitute into the simplified form: $3 + 1$. $\boxed{4}$

Watch for: 2 is what a mis-cancellation to $x - 1$ produces. The factor pair for -3 that sums to -2 is -3 and $+1$, so the surviving factor is $x + 1$; multiplying the factors back out is the check that catches it.

3. $\displaystyle\lim_{x\to 0} \frac{\sqrt{x+4}-2}{x}$.

Solution: Substitution gives $\dfrac{2-2}{0} = \dfrac{0}{0}$, so the expression must be rewritten. Multiply above and below by the conjugate $\sqrt{x+4}+2$:

$$\frac{\sqrt{x+4}-2}{x}\cdot\frac{\sqrt{x+4}+2}{\sqrt{x+4}+2} = \frac{(x+4)-4}{x\left(\sqrt{x+4}+2\right)} = \frac{1}{\sqrt{x+4}+2} \qquad (x \neq 0)$$

Substituting now is legal: $\dfrac{1}{\sqrt{4}+2} = \dfrac{1}{2+2}$. $\boxed{\frac{1}{4}}$

Watch for: $\frac{1}{2}$ comes from keeping only $\sqrt{x+4}$ in the new denominator and dropping the $+2$. The conjugate multiplication clears the radical from the *numerator*; the denominator keeps both terms.

4. $\displaystyle\lim_{x\to 2} \frac{1}{x-2}$.

The mistake: "the denominator shrinks to zero" fixes the size of the limit but says nothing about its sign, and the sign is different on the two sides.

Correct work: for x slightly less than 2, $x-2$ is a small negative number, so the quotient is large and negative:

$$\lim_{x\to 2^-} \frac{1}{x-2} = \boxed{-\infty}$$

For x slightly greater than 2, $x-2$ is small and positive:

$$\lim_{x\to 2^+} \frac{1}{x-2} = \boxed{+\infty}$$

The one-sided limits disagree, so the two-sided limit does not exist. Writing ∞ for it is a stronger claim than the function supports.

5. $\displaystyle\lim_{x\to 0} \frac{|x|}{x}$.

The mistake: $|x|$ and x do not cancel, because $|x|$ is x only when $x \geq 0$. For negative x it equals $-x$, which is where the sign flips.

Correct work: split at 0 using the definition of absolute value. For $x < 0$, $|x| = -x$, so the quotient is $\dfrac{-x}{x} = -1$ at every such x. $\boxed{-1}$

For $x > 0$, $|x| = x$, so the quotient is $\dfrac{x}{x} = 1$ throughout. $\boxed{1}$

The one-sided limits are different finite numbers, so the two-sided limit does not exist — a jump, not a blow-up. Note that the function never takes the value the student wrote at any negative x.

6. $\lim_{x \to 4} \frac{x-7}{x-4}$, both sides.
The mistake: "the right side is always positive infinity" ignores the numerator. Near $x = 4$ the numerator is close to $4 - 7 = -3$, a negative constant, so the sign of the quotient is the *opposite* of the sign of the denominator.
Correct work: approaching from the left, $x - 4$ is a small negative number, and a negative divided by a negative is positive:

$$\lim_{x \to 4^-} \frac{x-7}{x-4} = \boxed{+\infty}$$

Approaching from the right, $x - 4$ is small and positive, and a negative divided by a positive is negative:

$$\lim_{x \to 4^+} \frac{x-7}{x-4} = \boxed{-\infty}$$

The reliable procedure is two signs, then one division: sign of the numerator at the point, sign of the denominator on that side. The student's rule happens to be right only when the numerator is positive.

7. Continuity of the piecewise function at $x = 3$.
The mistake: $k = 14$ answers a different question. The number 14 is the value of the right-hand branch at $x = 3$; k is a constant sitting inside the left-hand branch, and it has to be chosen so the two branches meet.
Correct work: continuity at $x = 3$ requires the left-hand limit, the right-hand limit and the value to agree. The right-hand branch gives $5(3) - 1 = 14$. The left-hand branch gives $3^2 + k = 9 + k$. Setting them equal:

$$9 + k = 14$$

$$\boxed{k = 5}$$

With that k, both one-sided limits equal $9 + 5$. $\boxed{14}$
Watch for: taking $k = 14$ leaves the left-hand limit at $9 + 14 = 23$ while the right-hand limit is 14, so the graph jumps by 9 — the very gap the question asked to close.

8. Explain: what substitution reports. *(Open response — flagged for manual review; a model answer follows.)*

Model answer: Direct substitution is a test, not a method. When it returns a real number, the function is continuous at the point and the number is the limit. When it returns $0/0$, it has returned no information about the limit at all: the form is *indeterminate*, meaning limits of this shape can come out to any value or fail to exist. It is a signal to rewrite the expression — factor and cancel, multiply by a conjugate, or combine fractions over a common denominator — and then substitute into the rewritten form, which is legal because the two expressions agree everywhere except at the point itself, and a limit never looks at the point itself.

Nonzero over zero is a different report. The size of the quotient grows without bound, so the limit is infinite, but the statement is incomplete until the sign is settled. That requires looking at the sign the denominator carries on the side being approached, together with the sign of the numerator.

When the two one-sided limits are infinite with opposite signs, the function is not approaching any single behaviour, so the two-sided limit does not exist. Writing ∞ would claim the function grows without bound on both sides, which is a description of a different graph.

What a reviewer should look for: $0/0$ named as indeterminate rather than undefined, at least one algebraic route named as the next step, and a sign analysis that treats each side separately instead of assuming the right side gives $+\infty$.